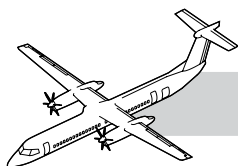


CHAPTER 56 WINDOWS

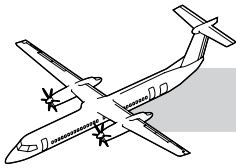
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CHAPTER 56 WINDOWS



56-00-00 INTRODUCTION

Refer to Figure 56-1. Windows

The Dash 8 Q400 Series aircraft has three classifications for windows that follow:

- Flight compartment windshields and side windows
- Passenger (compartment) cabin
- Door.

The flight compartment windows give the flight crew an adequate, unobstructed view outside the aircraft under all flight and weather conditions. The windows are structural elements which are able to withstand pressurization forces and impact from a bird strike. The passenger compartment windows give the passengers a view outside the aircraft. The windows are able to withstand pressurization forces and insulate the passengers from the extreme temperatures outside the aircraft.

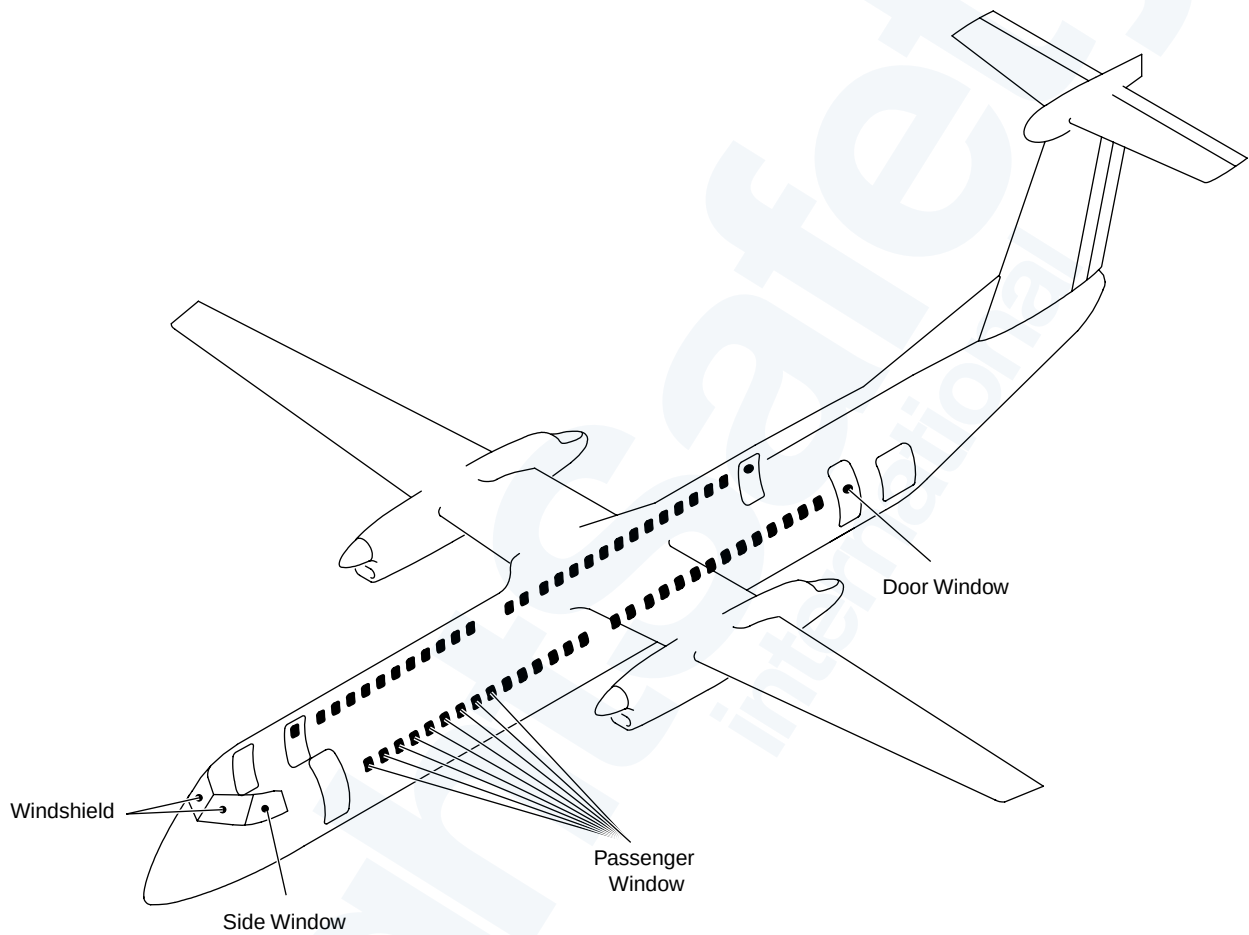
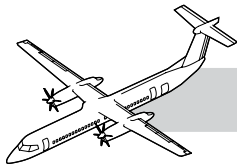
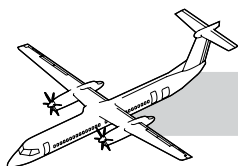
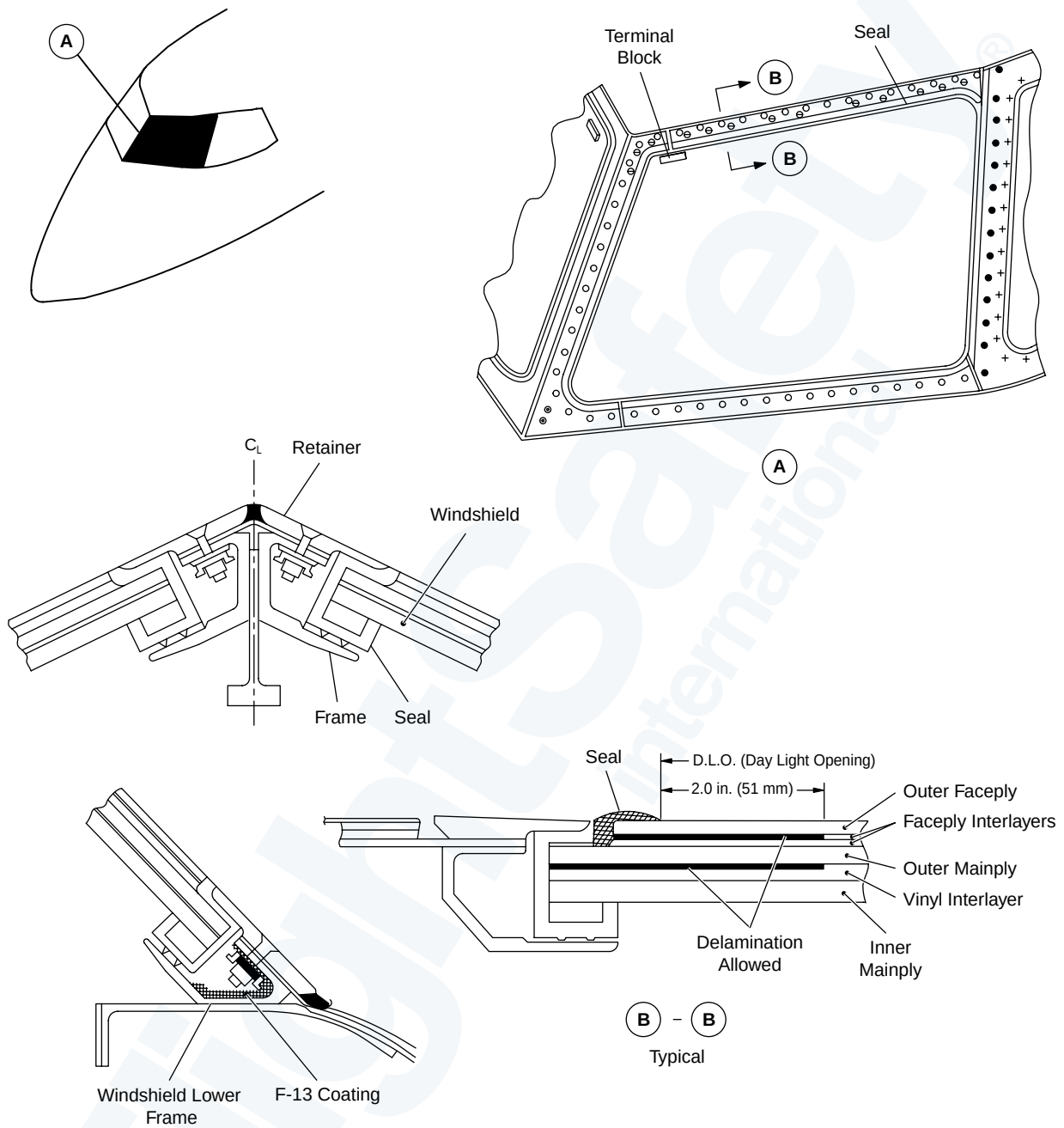
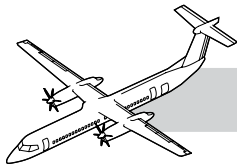


Figure 56-1. Windows

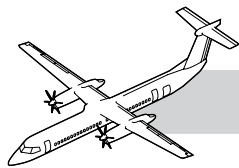


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**NOTE**

Total delamination not more than 2.0 inches (51mm) from Day Light Opening (DLO) of the seal.

Figure 56-2. Flight Compartment Windshield



GENERAL

Refer to Figure 56-2. Flight Compartment Windshield

The windshields are installed in the forward flight compartment. One is for the pilot; one for the co-pilot. The windshields are installed in an aluminium alloy window frame with four window retainers that are bolted to the frame. One retainer is also for the side window and is installed before the windshield is installed.

The windshields have panes that follow:

- Outer face ply
- Face ply interlayers
- Outer main ply
- Vinyl interlayer
- Inner main ply.

Each windshield has a thermally tempered glass outer face ply that is coated on the internal side with a conductive, metallic film. This supplies heating for anti-icing. The heating film covers most of the windshield area except for the forward section. Two urethane face ply interlayers are adjacent to the outer glass face ply.

The window also has two chemically strengthened glass main plies that are separated by a vinyl interlayer. A molded silicone rubber seal is bonded around the edge of the windshield for weather and pressure sealing. The two chemically strengthened glass main plies are the structural part of the windshield.

A terminal block and three temperature sensors are located near the top of the inboard side of the windows. The temperature sensors are located between the outer face and interlayer face plies.

The sensors give an interface between the windshields and the anti-icing system for window heating.

The windshield gives a satisfactory view of the aircraft for all flight and weather conditions. The windshields are structural parts that can withstand pressurization forces and an impact from a bird strike.

TRAINING INFORMATION POINT

The main types of damages that can occur to the windshields are as follows:

Scratches:

The three groups of scratches are:

- Faint scratches that can be seen but can not be felt with your fingernails
- Minor scratches that can be seen and felt with the fingernails < 0.004" (1.0mm)
- Major > 0.004" (1.0mm).

For details and depth measurement details please refer to the Aircraft Maintenance Manual.

Delaminating:

It is the separation of two or more layers of a laminated window panel. It usually occurs at the edges and/or at the corners of the window panel.

Cracks:

They are opening through all of the thickness of the window panels.

Bubbles:

Bubbles are trapped air within the interlayer usually circular in appearance.

Arcing:

It is an electrical discharge across a discontinuity in a wire, bus bar or electrical conductor.

Maintenance Practices/Ground Handling Requirements

As the inner pane of these windows have very little tolerance for damage, it is therefore very important to keep anything that can cause damage off the glare shield and away from the surface of the glass. This includes the journey logbook, pens, pencils, watches and other jewelry, and tools.

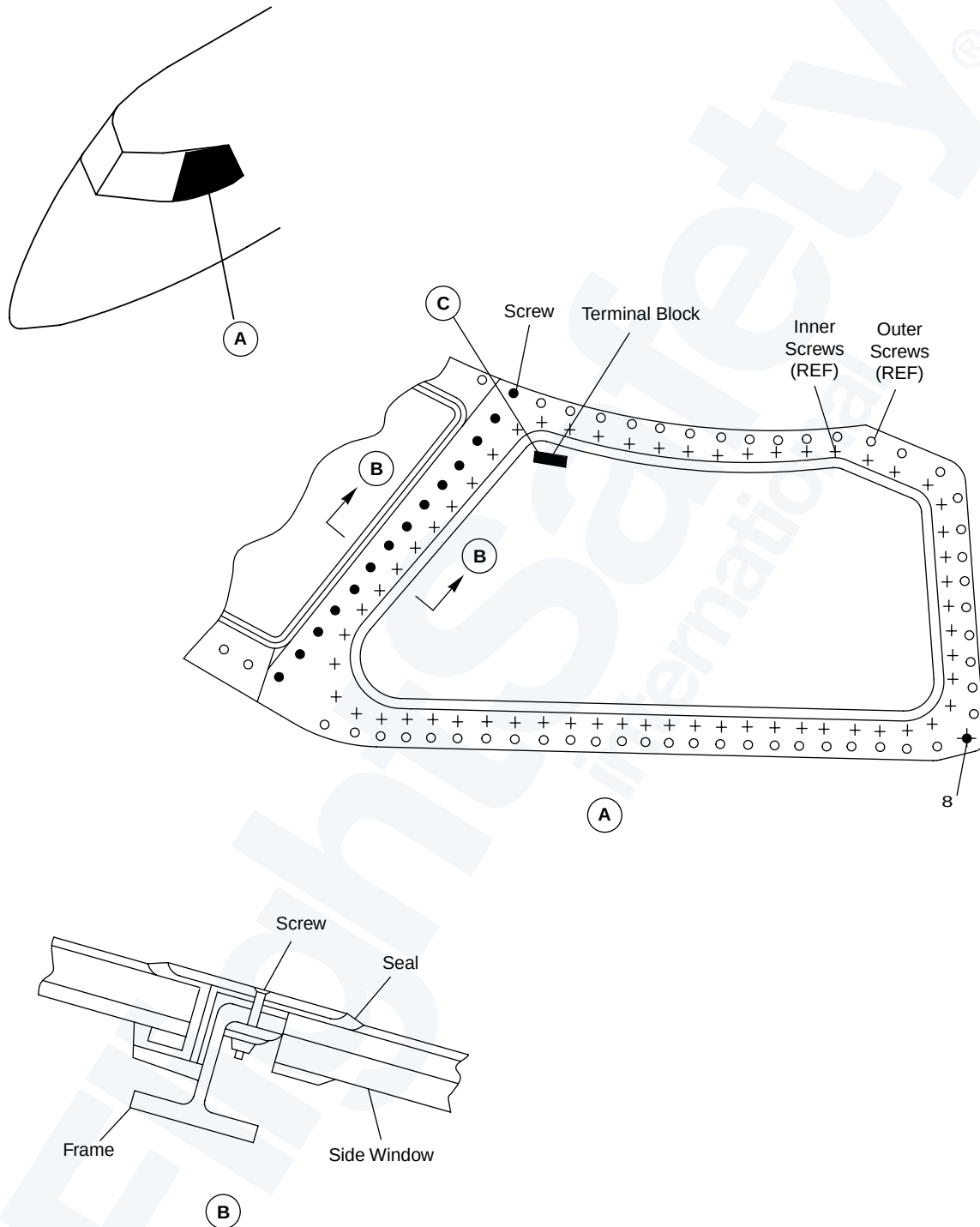
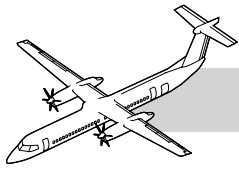
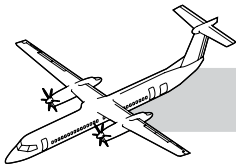


Figure 56-3. Flight Compartment Side Windows



Refer to Figure 56-3. Flight Compartment Side Windows

NOTE

The side windows are installed immediately aft of the pilot's and co-pilot's windshield. They are installed in an aluminium alloy window frame with a retainer that is bolted to the window and to the frame.

The pilot's and co-pilot's side windows are similar.

The pilot's side window has an outer, non-structural, acrylic face ply with a structural stretched acrylic main ply. The forward part of the face ply is coated on the internal side with a conductive, metallic film for anti-icing. There is a molded polysulfide weather seal and a rubber pressure seal around the edge of the window. A retainer is attached to the window with inner screws.

NOTE

The inner screws of the retainer are torqued by the manufacturer of the window and may be tightened only if air leakage occurs. If an inner screw is sheared, it must be replaced immediately. If more than five screws are sheared during one flight, the side window must be replaced.

A terminal block and three temperature sensors are located near the top of the inboard side of the pilot's side window.

They give an interface between the pilot's side windows and the anti-icing system for window heating.

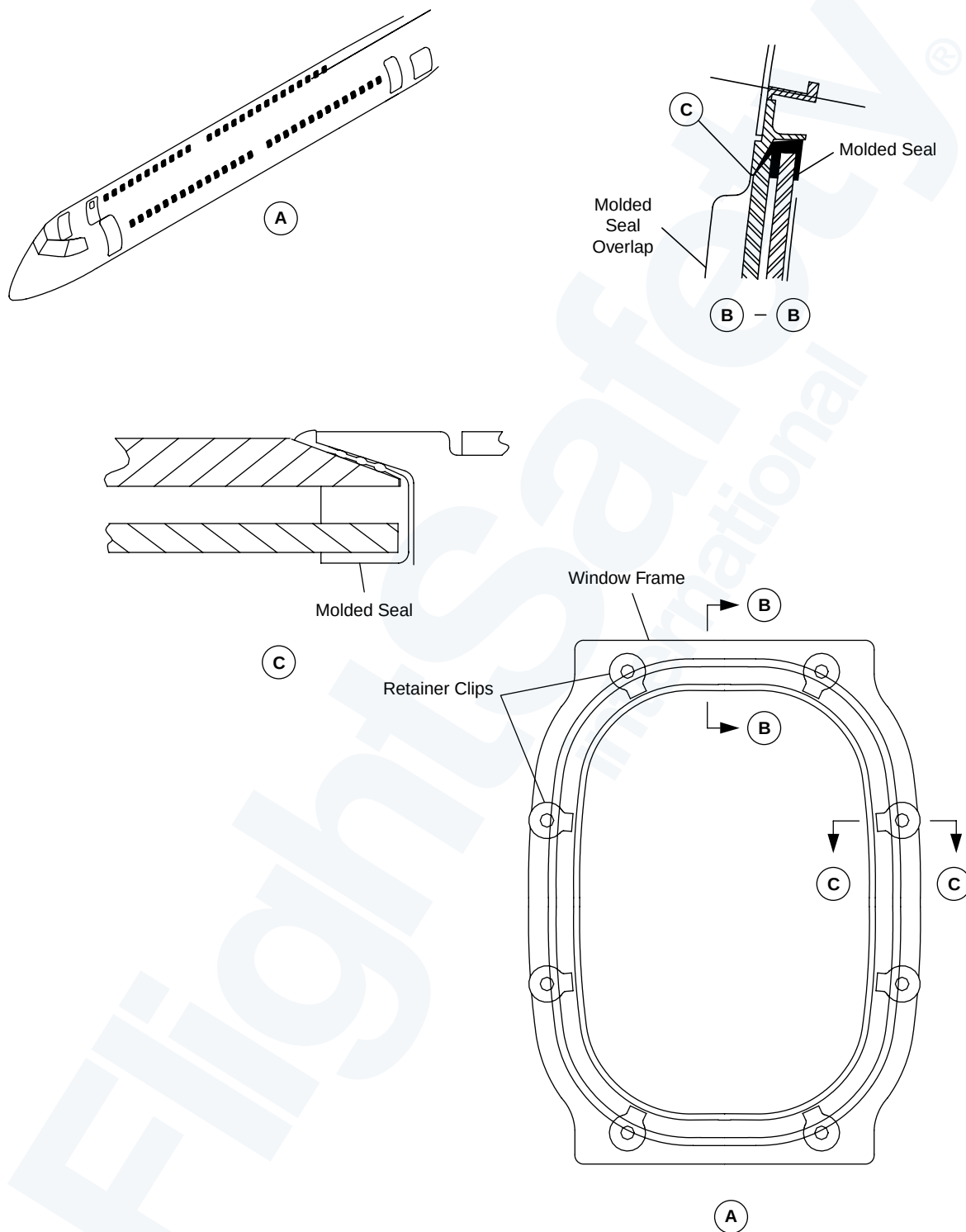
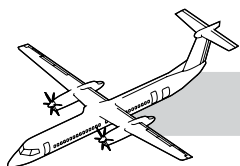
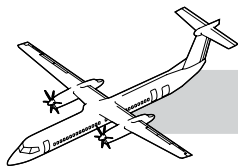


Figure 56-4. Passenger Compartment and Type II - Emergency-Exit-Door Windows



PASSENGER COMPARTMENT WINDOWS

Refer to:

- Figure 56-4. Passenger Compartment and Type II - Emergency-Exit-Door Windows
- Figure 56-5. Aft Passenger Door Window.

Windows are installed in each side of the passenger compartment and the doors that follow:

- Type II emergency-exit-door
- Aft passenger
- Service door.

Each window has an inner and outer acrylic pane in a molded silicone rubber seal.

The assembly is installed in an aluminium alloy window frame with retainer clips that are bolted to the frame.

The space between the panes of the passenger compartment windows are vented to the passenger compartment with two holes in the inner pane.

SPECIAL TOOLING

Commercially Available Wrench, Torque 0 to 40 lbf in (0 to 4.52 Nm).

SAFETY PRECAUTIONS

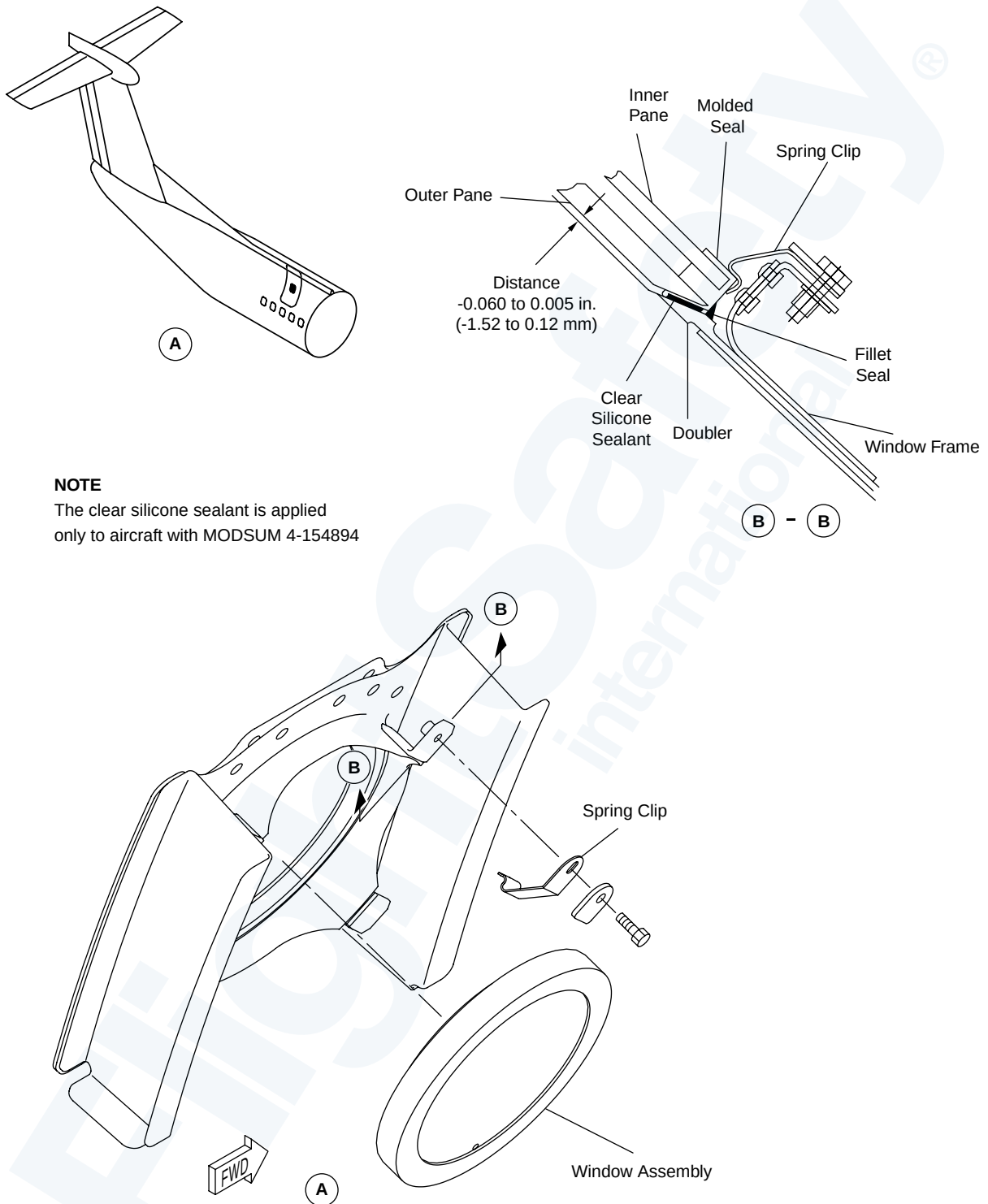
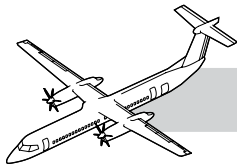
When you are near the windshield or side window, be very careful not to touch them with hard objects (tools, belt buckles, watches, jewellery). Hard objects can easily cause scratches or other damage to the windshield or side window.

When you remove or install the windshield or side window, make sure that you always put on cotton gloves. If you do not use cotton gloves you can cause scratches or damage to the windshield or side window. Scratches or damage can cause structural failure of the windshield or side window.

Do not use too much force or tools to remove the windshield. If you do, you can bend the retainer or cause damage to the frame or windshield.

Do not use lever bars or equivalent tools to set the windshield in the correct position. If you do, you can cause damage to the windshield.

Two people are required to remove and install the windshield in the aircraft structure.

**NOTE**

The clear silicone sealant is applied only to aircraft with MODSUM 4-154894

Figure 56-5. Aft Passenger Door Window